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The Editors
Information Processing & Management

Re: Submission of a Full Length Article — “Sieve: Training-Free, Model-Free, Question-Conditioned Context Compression for Efficient Long-Context LLM Inference: An Accuracy–Compression–Cost Frontier Study of Model-Free Information Selection”

Dear Editors,

Please consider the manuscript referenced above for publication in *Information Processing & Management* as a **Full Length Article**. I am the sole author of the manuscript and serve as its corresponding author.

The manuscript studies a question that sits between two well-mapped regions of the prompt-compression landscape. Neural compressors such as LLMingua and its successors recover most task quality at high ratios, but they require a hosted scorer, GPU capacity for at least one forward pass over the full context, and a tokenizer matched to the scorer. Truncation is free but question-agnostic, and it interacts badly with the positional biases of long-context models. Between the two lies a region that has not been mapped systematically: compressors that consult the question, cost milliseconds of arithmetic, and require no model at all. The manuscript measures how much quality that region gives up, at what compression, and at what computational cost.

The vehicle of the study is *Sieve*, a training-free, model-free sentence selector built from three corpus-local statistics: IDF-weighted coverage of query content terms, a U-shaped positional prior, and a redundancy penalty applied greedily under an explicit word budget, with selected sentences emitted in their original order. Its preprocessing is a linear-time pass over the context, its budgeted selection is $O(k n \bar{s})$, and on the evaluation container the full selector runs in tens of milliseconds on one CPU core (53.7 ms at a 6,000-word context), with no model parameters, no training, and no dependency beyond NumPy.

The evaluation is organized as a three-way accuracy–compression–cost frontier study over three evidence-annotated benchmarks — QASPER, HotpotQA (distractor setting), and 2WikiMultihopQA — totaling 1,600 records under a reader-free evidence-recall protocol, plus end-task question answering with a production reader at the $4\times$ operating point, with sample sizes stated and claims scoped accordingly. The main findings, all reported with paired bootstrap intervals, are: (i) Sieve retains more gold evidence than every question-agnostic baseline on all three datasets, with margins over head truncation of 12.8, 44.8, and 22.8 recall points at $4\times$; (ii) a BM25 sentence-ranking baseline, which the manuscript adds as the strongest question-conditioned model-free comparator, exceeds the full selector in raw recall on all three datasets — a result the manuscript reports as measured and explains through the selector’s own ablations as the positional prior trading recall for placement where readers attend reliably; (iii) a GPT-2 perplexity gate, measured on the same hardware as every other

method, reaches 28.7% evidence recall at $4\times$ on QASPER at $255\text{--}320\times$ Sieve’s measured cost; and (iv) evidence recall correlates only weakly with end-task answer F1 (pooled Pearson 0.30), so the two protocols are reported separately throughout. A formal analysis states the cost of each pipeline stage and the selection behavior under an explicit lexical-coverage condition, and a robustness study covers evidence placement, hyperparameter sensitivity, and cross-dataset transfer without retuning.

I believe the manuscript fits the scope of *Information Processing & Management*. It is a quantitative study of question-conditioned information selection: it builds directly on classical information-retrieval machinery (IDF weighting, BM25, maximal marginal relevance), evaluates selection quality with recall protocols over annotated evidence, and connects selection decisions to the processing and cost of managing long contexts in information systems. The frontier analysis reports quality, compression, and compressor cost for every method, giving practitioners a measured basis for deciding when statistical selection suffices and when a hosted scorer earns its infrastructure.

The manuscript substantially extends an earlier conference-format study of the same method, as described in the Introduction, with a three-dataset evaluation, a new BM25 baseline, a same-hardware measured comparison against a neural perplexity gate, formal analysis, and a recall–accuracy correlation study.

I confirm that: the manuscript is original and has not been published previously in any form; it is not under consideration by any other journal; I have no known competing financial interests or personal relationships that could have appeared to influence the work reported; and the work received no specific funding. The benchmarks used are public (QASPER, HotpotQA, 2WikiMultihopQA), and the released repository reproduces every table and figure from raw checkpoints.

Thank you for considering the manuscript. I look forward to the reviewers’ comments.

Sincerely,

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